

Week 1 – LLM Model Selection for Patient Story Generation

Project: LLM module for patient history → clinician summary + patient story

Week: 1 – Model Selection (Review Gate #1)

Objective

Select a primary and fallback LLM API for generating de-identified patient summaries and patient-friendly narratives from structured EHR-style data and opioid-risk information. We only send de-identified patient histories to the LLM, meaning the model sees medical details (conditions, treatments, labs, etc.) but no information that can directly identify an individual patient.

Candidate APIs considered

- OpenAI: GPT-4.1, GPT-5, GPT-5.1
- Anthropic: Claude Sonnet 4.5 (with Haiku / Opus noted as variants)
- Google: Gemini 1.5 / 2.0 / 3 Pro (considered but not selected for initial deployment)

Compared on price per 1K tokens (input/output), context length, safety posture, availability, SDK/docs, and overall fit for the “patient story” use case.

Selected models

Primary: OpenAI GPT-4.1

- Context: ~1,000,000 tokens
- Pricing (approx., per 1M tokens):
 - Input: \$2 → \$0.002 / 1K
 - Output: \$8 → \$0.008 / 1K
- Fit:
 - Very strong summarization and explanation quality.
 - Huge context window supports full longitudinal histories + templates.
 - Stable, well-documented API with good Python support.

Estimated monthly cost (3,000 summaries/month)

- Input: 1,500 tokens/story → 4.5M tokens/month
- Output: 1,000 tokens/story → 3.0M tokens/month
- Cost $\approx 4.5\text{M} \times \$2/1\text{M} + 3.0\text{M} \times \$8/1\text{M} \approx \$33/\text{month}$

Fallback: Anthropic Claude Sonnet 4.5

- Context: up to ~1,000,000 tokens (long-context tier)

- Pricing (approx., per 1M tokens):
 - Input: \$3 → \$0.003 / 1K
 - Output: \$15 → \$0.015 / 1K
- Fit:
 - Strong long-context summarization; conservative, safety-focused behavior.
 - Provides vendor diversity (Anthropic API / Bedrock / Vertex) if OpenAI access is limited.

Estimated monthly cost (same workload)

- Input: $4.5\text{M} \times \$3 / 1\text{M} \approx \13.5
- Output: $3.0\text{M} \times \$15 / 1\text{M} \approx \45
- Total $\approx \$59/\text{month}$

Why not GPT-5 / GPT-5.1 / Gemini as primary?

- **GPT-5 / GPT-5.1** – excellent for complex reasoning and agentic behavior, but:
 - Smaller context ($\approx 270\text{--}400\text{k}$) than GPT-4.1's 1M.
 - Higher output price ($\sim \$10 / 1\text{M}$ tokens).
 - Their strengths (tool use, planning) are not critical for this first-phase narrative generator.
- **Gemini Pro / 3 Pro** – strong multimodal models but:
 - Require extra infrastructure (Google Cloud / Vertex).
 - Similar or higher costs, with no clear advantage for simple text summarization.

Key risks and mitigations

- **Hallucination / incorrect clinical statements**
 - Use only de-identified data; all outputs manually reviewed by clinicians / TAs.
 - Prompting will explicitly instruct the model not to invent new diagnoses or medications.
- **Vendor lock-in / outages**
 - Implement a thin LLM wrapper so we can switch between GPT-4.1 and Claude Sonnet 4.5 via configuration.
- **Cost overruns**
 - Start with small test batches and monitor token usage.
 - If volume grows, consider cheaper variants (Claude Haiku 4.5, Gemini Flash) for bulk drafts.

Recommendation:

Adopt GPT-4.1 as the primary model for patient story generation and configure Claude Sonnet 4.5 as the fallback provider. This combination offers long context, strong summarization quality, reasonable cost, good safety characteristics, and flexibility to adopt newer models later without changing the overall architecture.

B. Detailed Week 1 Write-up (Model Survey & Rationale)

1. Project context

The goal of this project is to build an LLM-based module that can take de-identified patient histories and opioid-risk information and turn them into:

1. A clinician-oriented summary (concise, structured, clinical language), and
2. A patient-friendly story (plain-language explanation of what is going on and why).

We only send de-identified patient histories to the LLM, meaning the model sees medical details (conditions, treatments, labs, etc.) but no information that can directly identify an individual patient. This story-generation component will sit behind an API and eventually integrate with the model server, dashboards, and any patient-education interface we build later in the semester. For Week 1, the focus is only on selecting and justifying the LLMs I want to use.

2. Requirements for the model

Before picking models, I wrote down what this system actually needs:

• Strong summarization and explanation

- Compress structured diagnoses, medications, encounters, and risk scores into a coherent narrative.
- Translate clinical language into understandable layperson language without losing meaning.

• Long-context support

- Able to handle longitudinal EHR-like histories, plus prompt templates and risk explanation text.
- Target: hundreds of thousands of tokens, ideally close to 1M tokens to stay future-proof.

• Reasonable cost at the expected scale

- Rough assumption: ~100 stories/day → ~3,000 stories/month.

- Each story call $\approx 1,500$ input tokens + 1,000 output tokens (two narratives: clinician + patient).
- Aim: keep spend in the tens of dollars per month, not hundreds.
- **Safety and workflow fit**
 - Only de-identified data is sent; the LLM is purely a documentation and decision-support tool.
 - A human (clinician / TA) must review and approve all outputs.
- **Practical integration**
 - Good Python SDKs and stable APIs.
 - Clear docs and minimal surprises during the semester.

For this stage, I do not need a full agent that calls tools or browses the web. The milestone is simple: “input JSON patient history -> output two narratives.”

3. Candidate models

I looked at three main families: OpenAI GPT, Anthropic Claude, and Google Gemini. Within each family I considered both the “stable workhorse” models and the newest flagship / throughput variants.

3.1 OpenAI – GPT-4.1, GPT-5, GPT-5.1

GPT-4.1 (gpt-4.1)

- Context: $\sim 1,000,000$ tokens
- Max output: $\sim 32k$ tokens
- Pricing (rough, per 1M tokens): Input $\approx \$2$, Output $\approx \$8$
- Characteristics:
 - “Smartest non-reasoning” model in the 4-series.
 - Very strong at summarization, explanation, and long-context tasks.
 - No explicit extra “thinking step,” so latency and cost are predictable.
 - Natural fit for turning long structured patient histories into narratives.

GPT-5 (gpt-5-chat-latest, gpt-5-mini, etc.)

- Context: $\approx 400k$ tokens, $\sim 128k$ output
- Pricing: $\approx \$1.25$ / 1M input, $\approx \$10$ / 1M output
- Characteristics:

- New 5-series model with stronger reasoning and multimodal support.
- Better suited for agents, complex workflows, and heavy tool use.

GPT-5.1 (gpt-5.1-chat-latest)

- Context: $\approx 270\text{--}400\text{k}$ tokens (variant-dependent)
- Pricing: similar to GPT-5 ($\approx \$1.25/1\text{M}$ input, $\$10/1\text{M}$ output)
- Characteristics:
 - OpenAI's flagship reasoning model.
 - Adds a more explicit “thinking” phase and better tool-calling.
 - Ideal when complex multi-step reasoning or guideline lookup is required.

Fit for this project

All three can produce good clinical and patient-facing narratives. The key differences:

- GPT-4.1 is the only one with a 1M-token context.
- GPT-4.1 has lower output cost than 5/5.1.
- GPT-4.1 is already widely used for long-context summarization.

For the current summarization + explanation scope (not complex agents), GPT-4.1 is the best match among OpenAI models.

3.2 Anthropic – Claude Sonnet / Haiku / Opus (Claude 4 family)

Claude Sonnet 4.5 (claude-sonnet-4-5)

- Context: up to $\sim 1,000,000$ tokens (long-context configs)
- Pricing (rough, per 1M): Input $\approx \$3$, Output $\approx \$15$
- Characteristics:
 - “Balanced” Claude 4 model.
 - Strong long-context writing and reasoning.
 - Emphasizes safety and Constitutional AI, good for sensitive health related text.

Claude Haiku 4.5 (claude-haiku-4-5)

- Context: large (hundreds of thousands of tokens), smaller/cheaper than Sonnet
- Characteristics:
 - Small, fast, low-cost model.
 - Similar capability to earlier Sonnet models; optimized for latency and cost.

Claude Opus 4 (claude-opus-4)

- Context: large (but not usually 1M)
- Characteristics:
 - Anthropic's most powerful model; targeted at very hard reasoning and coding tasks.
 - Also the most expensive in the Claude 4 line.

Fit for this project

- Claude Sonnet 4.5: closest counterpart to GPT-4.1 (long context, strong summarization, strong safety posture).
- Haiku 4.5: attractive if I later need to generate very large volumes of stories cheaply.
- Opus 4: overkill for routine narrative generation.

So Claude Sonnet 4.5 is the main Anthropic model I plan to support as fallback.

3.3 Google Gemini – 1.5 Pro / Flash, 2.0 Flash, 3 Pro

Gemini 1.5 Pro (gemini-1.5-pro)

- Context: up to ~128k tokens standard (with extended tiers)
- Pricing: flagship-level (similar to other high-end models)
- Characteristics:
 - Strong multimodal and reasoning capabilities.
 - Good for complex analysis, code, and mixed text-image-video tasks.

Gemini 1.5 Flash / Flash-8B (gemini-1.5-flash)

- Context: similar tiers to Pro
- Pricing: very cheap; optimized for throughput
- Characteristics:
 - Extremely fast and inexpensive.
 - Ideal for bulk summarization where small quality trade-offs are acceptable.

Gemini 2.0 Flash & Gemini 3 Pro

- 2.0 Flash: new generation of ultra-cheap, high-throughput models.
- 3 Pro: Google's latest flagship reasoning/multimodal model.
- Both are promising for future, but add infra complexity (Google Cloud / Vertex).

Fit for this project

Gemini models are appealing for future multimodal work, but for this semester:

- They require extra setup.
- Flagship models don't give a clear advantage for plain-text summarization.
- Starting with OpenAI + Anthropic is simpler and fits the tools I'm already using.

4. Why I'm not leading with GPT-5 / GPT-5.1 / highest-end models

Main question: Do I need a flagship reasoning agent right now, or just a very good summarizer with lots of context?

For this stage:

- The core task is summarization + explanation, not complex multi-tool reasoning.
- GPT-4.1 and Claude Sonnet 4.5 already provide:
 - Very large context windows (up to ~1M tokens),
 - Strong summarization quality,
 - Manageable cost.

Models like GPT-5/5.1, Claude Opus, and Gemini 3 Pro are great for advanced agents and multimodal reasoning, but they are more expensive and their extra capabilities are not fully used in the initial "patient history → two narratives" workflow. So I treat them as future upgrade options, not the default choice.

5. Final selection and rationale

- Primary model: OpenAI GPT-4.1
 - 1M-token context, strong summarization and explanation, stable API, reasonable cost.
- Secondary model: Anthropic Claude Sonnet 4.5
 - Similarly large context, strong safety posture, vendor diversity, good for A/B comparisons.

6. Real-world usage (brief reference)

Similar GPT-4-class and Claude Sonnet-class models are already being used for clinical note and patient-data summarization by:

- **Carbon Health**
- **Nuance (Dragon Ambient eXperience / DAX Copilot)**
- **Abridge**
- **BastionGPT / Elion**
- **Hathr.AI**

and in research published in:

- JAMA Network Open
- Scientific Reports
- Acta Orthopaedica
- CL4Health (LREC-COLING)
- Frontiers in Digital Health

This gives additional confidence that the chosen model families are appropriate for patient history summarization.

7. One-sentence summary

For Week 1, my plan is to implement and test the patient story generation API primarily with GPT-4.1, with Claude Sonnet 4.5 as a secondary model, because together they provide long context, strong summarization quality, reasonable cost, good safety, and an easy path to future experiments without changing the overall architecture.